

PRAVEEN MANDALA

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PROFESSIONAL SUMMARY

Computer Science Student with a strong foundation in system design and software architecture, demonstrated through hands-on projects in scalable application development. Passionate about solving complex engineering challenges with an emphasis on clean code, performance optimisation, and maintainability. Proven ability to design, build, and deploy end-to-end solutions, from conceptualisation to implementation.

Seeking to contribute strong foundational design skills and a hands-on approach to a forward-thinking engineering team, with a focus on developing scalable and impactful software.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

(2023 – 2027)

Aditya College of Engineering & Technology, Surampalem

CGPA: 8.79/10

INTERNSHIP & EXPERIENCE

FULL STACK DEVELOPER INTERN - Zengen Technologies

(May 2025 – Aug 2025)

- Contributed to the development of backend services, gaining hands-on experience in writing clean, maintainable code and understanding principles of scalability within a live application environment.
- Assisted in resolving functional issues across the stack, learning to debug code, analyse logs, and apply fixes under guidance to improve system stability.
- Supported the implementation of new features by developing frontend components and integrating them with backend APIs, gaining practical exposure to the full software development lifecycle in a team setting.

PROJECTS

LLM Evaluation Framework [AI Research]

[Link](#)

- LLM as a Judge testing suite to quantitatively benchmark RAG pipelines on metrics like Faithfulness and Relevance.
- Created automated benchmark pipelines to validate model performance across multiple providers (OpenAI, Gemini).
- A scalable evaluation system by leveraging the factory pattern, designing a flexible API framework that enables seamless integration and swapping of various LLM providers.

Multi-Modal AI Vision Platform [SaaS Application]

[Link](#)

- Building an AI SaaS Visual Question Answer(VQA & OCR) by secure AWS cloud infrastructure (S3, EC2) and building the application with Next.js.
- Implemented a secure media pipeline. Pre-signed URLs for temporary, controlled access to user assets.
- Designed a responsive frontend with real-time analysis rendering and drag-and-drop file handling.

Real-Time Feature Store [MLOps infrastructure]

[Link](#)

A low-latency features serving system for ML model using FastAPI and Faust agents for stream processing.

- Dual-storage strategy with Redis for Sub-millisecond retrieval and PostgreSQL for history consistency.
- Implemented asynchronous data synchronisation to ensure zero-latency impact on inference requests.

Secure E2E Encrypted Chat [Security System]

[Link](#)

- A privacy-focused messaging architecture featuring End-to-End Encryption (E2EE) using WebCrypto API.
- Used ECDH (Elliptic Curve Diffie-Hellman) for secure key exchange and AES-GCM for client-side encryption.
- A resilient presence system using Socket.io and Redis to handle real-time user status updates at scale.

SKILLS

- Languages:** Java, C++, Python, JavaScript
- Backend & Systems:** Node.js, FastAPI, GenAI, Microservices, REST APIs, WebSockets, Kafka, Redis, NGINX
- Core CS:** Data Structures & Algorithms, OOP, OS, DBMS, Computer Networks, Cryptography
- Cloud & DevOps:** AWS(S3, EC2), Docker, Kubernetes, CI/CD Pipelines
- Frontend:** React.js, Next.js, Tailwind CSS, HTML5
- Databases:** PostgreSQL, MySQL, MongoDB
- Tools:** Git, Postman, Figma, GitHub

CODING PROFILES & CERTIFICATIONS

- Cisco C Essentials
- Top badges in MySQL, Java, Python and JavaScript on HackerRank.
- 200+ problems solved across Leetcode & GeeksForGeeks.